A Component-Level Audit of No-Reference Quality Scoring in a Multi-Video Summarization Pipeline

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Abstract

A visual-quality criterion in a multi-video summarization pipeline is implemented with Bag-of-Visual-Words and term frequency–inverse document frequency, techniques designed for retrieval rather than perceptual assessment. This study audits that operation while leaving the remaining pipeline stages unchanged. An exact factorization separates descriptor count from mean IDF weight; measurements on three image-quality databases and an ablation of 19 extractors show that descriptor count dominates the evaluated configurations. On 800 paired decisions constructed by degrading the same broadcast content, the deployed score selects the unmodified member in 65.1% of cases and reverses direction under white noise and JPEG compression; LIQE reaches 99.5%. A dependency-aware screening protocol retains LIQE, QualiCLIP and CLIP-IQA+. The downstream technical output of DOVER, reserved from screening, reaches 97.6% accuracy on the constructed pairs ($[\mathbf{96.1, 99.0}]$). With the same ten frames and mean pooling, SIFT+BoVW and LIQE still differ in 16 of 42 clusters, with a positive hierarchical DOVER contrast. Replacing the complete deployed operation, while holding all other pipeline stages fixed, changes 22 of 42 segments. The DOVER technical-branch point estimate rises by 25.5 percentage points of the global achievable range, but its topic-block $[\mathbf{-12.1, +54.9}]$ and hierarchical $[\mathbf{-17.2, +59.6}]$ intervals both cross zero. The application therefore establishes changed decisions, not improved viewer experience or a confirmed population-level output effect; no human compositional preferences were collected.

Keywords: Multi-video summarization, No-reference image quality assessment, Content selection criteria, Benchmark evaluation, Broadcast news video

1 Introduction

Video summarization reduces long audiovisual collections to selected keyframes, shots or segments [3, 34]. When several sources cover the same topic, selection must also handle cross-source redundancy [21, 26, 40]. A multi-video pipeline therefore combines several criteria: a segment can be relevant yet redundant, visually informative yet technically poor, or temporally misplaced.

HSMVideoSumm makes those criteria inspectable. A study with human summarizers of multi-source news identified five recurrent strategies—removing introductory passages, filtering subjective content, detecting redundancy, selecting for visual quality and ordering chronologically—and implemented them as separate stages [4, 5]. The present study concerns only the fourth stage. Segmentation, subjectivity filtering, redundancy detection and ordering remain unchanged. Once the pipeline treats segments as a redundant cluster, visual quality acts as a tie-breaker; this conditional design does not imply that cluster members have identical semantic utility.

The scientific issue is the distinction between a criterion and the operation that implements it. The criterion is human-grounded because it was observed in the summarization study. Its implementation is a later engineering choice: Scale-Invariant Feature Transform (SIFT) descriptors are quantized through Bag-of-Visual-Words (BoVW), weighted with term frequency–inverse document frequency (TF-IDF), and summed [10, 23, 30, 31]. SIFT, BoVW and TF-IDF were developed for matching, retrieval and categorization, not for predicting perceptual quality. Calling their aggregate a quality score does not establish construct validity.

No-reference image quality assessment (NR-IQA) provides alternatives because segments from different broadcasters have no common undistorted reference. The available families include natural-scene-statistics, supervised convolutional and transformer regressors, and vision–language models [1, 19, 24, 25, 45]. The question is not whether a newer model produces a higher number, but whether the deployed operation responds to the construct it names, whether that behaviour can be explained, and whether replacing the operation changes the pipeline output.

The study makes three contributions. First, it derives an exact factorization that isolates the descriptor-count and mean-IDF-weight channels of the deployed score, then measures which channel dominates across image databases, broadcast segments and 19 convolutional neural network (CNN) extractor substitutions. Second, it combines human-score association with within-reference degradation sensitivity and a constructed experiment containing 800 paired decisions between an unmodified signal and a deterministically degraded counterpart. Third, it applies a common screening rule and substitutes the complete visual-quality operation in 42 redundancy clusters while holding every other pipeline stage fixed, measuring both changed selections and model-reference scores of the selected segments. Training-database overlap, detectable-keypoint conditioning, alternative pooling, overlay strata, predictions recorded in a commit whose metadata predates the experiment and computational details are auxiliary diagnostics reported in the supplementary material.

The resulting claim is deliberately bounded: the factorization explains the observed count dominance and failure of the evaluated BoVW+TF-IDF configurations. It is not a theorem that every possible visual vocabulary, normalization or extractor must

fail. Likewise, the application study separates Q-Align, which defines the in-domain screening floor, from DOVER, which is used only for output sensitivity; neither model substitutes for human preference judgments.

2 Related work

2.1 Quality-aware multi-video summarization

Extractive summarization selects existing frames or temporal segments, whereas abstractive approaches generate derived representations [3, 34]. Multi-video methods add explicit cross-source redundancy and diversity. Video-MMR formalized the setting [21]; sparse and collaborative formulations improved diversity across topic-related videos [26, 27]; graph models subsequently represented inter-video relations directly [40]. These methods address representativeness, coverage and redundancy, but they do not validate a visual-quality proxy used to choose between redundant candidates.

HSMVideoSumm instead derived selection criteria from human summarization practice and implemented each criterion as an explicit pipeline stage [4, 5]. This modularity permits a component-level audit: the same clusters, filters and ordering can be held fixed while the operation used for visual quality is tested and replaced. The object of comparison is therefore neither a new summarizer nor an overall benchmark against SumMe or TVSum; it is one inspectable decision inside a multi-source news pipeline.

The use of image-quality assessment in summarization is not itself new. VSQUAL used objective IQA measures for shot-boundary detection, keyframe extraction and summary construction [9]. Jeong et al. also combined representativeness with contrast, blur and image-skew measures when selecting keyframes for static video summaries [18]. The present contribution is different: it does not claim novelty for adding IQA, but for a mechanistic construct-validation audit of an already deployed quality operation, followed by dependency-aware propagation of an operation-level substitution.

2.2 No-reference quality assessment and transfer

Image quality assessment predicts perceived degradation. Full-reference measures require an undistorted original and are unsuitable when candidates come from different broadcasters [37, 43]. NR-IQA instead estimates quality from the test image alone. BRISQUE and NIQE model departures from natural-scene statistics [24, 25]; DBCNN and HyperIQA learn opinion-supervised convolutional representations [33, 44]; MUSIQ and TOPIQ use multi-scale or top-down attention [8, 19].

Vision–language formulations based on Contrastive Language–Image Pre-training (CLIP) provide a different inductive bias. CLIP-IQA uses antonym prompts [35]; LIQE jointly predicts quality, distortion and scene [45]; QualiCLIP learns from rankings of synthetically degraded pairs without opinion scores [1]. Q-Align maps multimodal-model outputs to discrete quality levels [39]. These models differ not only architecturally but also in their exposure to opinion databases. A result on training data measures fit, whereas use in a new broadcast domain requires transfer.

Cross-database evaluation is therefore essential, and published work has shown that in-database NR-IQA performance can overstate generalization [6, 28].

KADID-10k supplies controlled synthetic distortions [22]; KonIQ-10k and LIVE In the Wild (CLIVE) contain authentic in-the-wild images [12, 14]. Their differences make them complementary validation settings, but comparisons among unmatched databases and checkpoints do not identify causal effects of training exposure or distortion regime. The complete descriptive contrasts are consequently kept in Supplementary Table S15 rather than presented as a principal contribution.

Two further cautions affect the broadcast application. First, professional and user-generated video can contain temporal impairments that a frame-wise image score cannot observe; video-quality models were developed partly for this reason [15, 36]. Second, high correlation on a reused database need not imply difficult pairwise discrimination. Database range and subjective-score uncertainty affect correlation [20], and recent multimodal benchmarks report that comparative quality judgments can behave differently from absolute scoring [46]. Newer distributional quality models also exceed earlier reference-model results on some databases [42]. Accordingly, Q-Align and DOVER are treated here as complementary contemporary references, not definitive ground truth or evidence that temporal modeling itself causes an advantage.

2.3 The unresolved measurement problem

SIFT, BoVW and TF-IDF encode local-feature occurrence and rarity for retrieval [10, 23, 30, 31]. Spatial Information similarly quantifies scene complexity rather than perceptual quality [17, 29]. Such features may correlate with opinion scores in a particular collection, but correlation does not show that they measure quality or remain valid under controlled degradation.

The gap addressed here is narrower than the general problem of quality-aware summarization. Existing work establishes multi-video selection criteria, specialized NR-IQA methods and prior use of IQA in summarization, yet the operation implementing HSMVideoSumm's human-grounded visual-quality criterion was not validated as a quality measure. This study links a mechanistic analysis of that operation to external human scores, construction-defined degradation pairs and a controlled operation-level substitution in the pipeline. That combination distinguishes construct validation from merely integrating an available IQA model.

3 Audit design

The audit combines four forms of evidence (Fig. 1): a mechanistic analysis of the deployed score, external validation against human opinion scores, construction-defined degradation pairs in the broadcast domain, and an operation-level substitution in the summarization pipeline. A method is eligible only if it responds to controlled degradation, generalizes to every unseen opinion database and exceeds a reference-model agreement floor in the deployment domain.

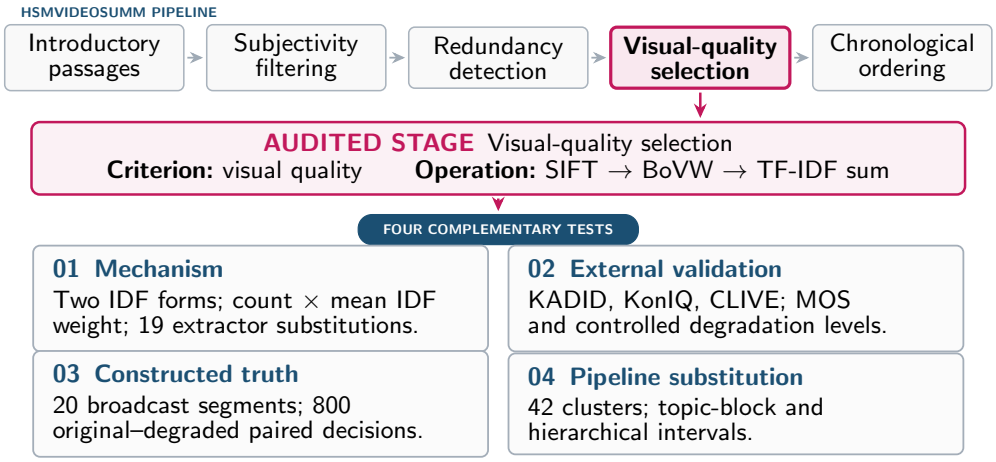


Fig. 1 Scope and evidence flow. Top: the five HSMVideoSumm stages, with visual-quality selection highlighted. Bottom: four complementary tests of the operation implementing that criterion. All other pipeline stages remain fixed.

3.1 Data and units of analysis

The external benchmark uses KADID-10k, KonIQ-10k and LIVE In the Wild (CLIVE). KADID-10k contains 10,125 distorted images derived from 81 references, with 25 distortion types at five levels and Differential Mean Opinion Scores (DMOS) [22]. KonIQ-10k contains 10,073 authentically distorted images, approximately 1.2 million ratings and Mean Opinion Scores (MOS) [14]; CLIVE contains 1,162 mobile photographs with authentic distortions [12]. No method is retrained or fine-tuned.

The deployment analysis uses the redundancy stage produced by HSMVideoSumm over 19 of its 21 news topics: 42 multi-member clusters, 93 segment slots and 83 distinct segments. The other two topics contain no multi-member cluster. These clusters are operational inputs, not proof of equal semantic utility. The original human redundancy annotations cover only 15 clusters and are retained as a supplementary sensitivity analysis (Tables S1, S7, S8 and S11); the 42-cluster set is primary because it is the set encountered by the deployed pipeline.

Still-image scorers receive frames sampled uniformly at 1 frame per second and are pooled by the arithmetic mean,

$$S(s) = \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(f_i), \tag{1}$$

where ϕ is the per-frame score. The deployed operation instead applies its own keyframe selector, quantizes retained descriptors into a vocabulary of $K = 300$ visual words and sums

$$S_{\mathrm{BoVW}}(s) = \sum_{i} \sum_{v=1}^{K} \mathrm{tf}(v, f_i)\, \mathrm{idf}(v). \tag{2}$$

Sampling, scoring and pooling jointly define the operation occupying the visual-quality stage, so the primary comparison evaluates complete operations while holding every other stage fixed. A matched-input control additionally gives SIFT+BoVW and LIQE the same fixed set of at most ten uniformly spaced 1-fps frames and arithmetic-mean pooling. For SIFT, common-frame descriptors fit a seeded per-cluster vocabulary with $K_{\text{eff}} = \min(300, n_{\text{descriptors}})$; candidate segments are the documents for the deployed $\log_{10}(K_{\text{eff}}/\text{df})$ IDF. Empty-descriptor frames remain as zeros in the mean; all observed segments yielded descriptors and no top-score ties occurred. This isolates scorer identity within the matched-input regime (Tables S6, S9 and S11).

3.2 Scorers, references and polarity

The complete-case comparison contains the deployed SIFT+BoVW score and ten NR-IQA candidates: BRISQUE, NIQE, DBCNN, HyperIQA, MUSIQ, TOPIQ, CLIP-IQA, CLIP-IQA+, LIQE and QualiCLIP. Keypoint density, Spatial Information, deep feature norms and the second IDF formulation are diagnostic controls. A separate ablation replaces SIFT with 19 convolutional extractors while holding the vocabulary, weighting and aggregation fixed. ARNIQA and TReS extend coverage on completed axes but are excluded from complete-case screening; MANIQA was not completed before the experimental freeze (Table S13). Released-weight supervision, checkpoint role, polarity and GPU requirements are summarized in Table 1 and listed in full in Table S13 [2, 13, 41].

Q-Align is an in-domain reference model rather than a candidate [39]. The `q-future/one-align` checkpoint is run with the `quality` task, `image` input, 4-bit NormalFloat4 (NF4) double quantization and FP16 compute, then pooled by Eq. 1. It was among the strongest public NR-IQA models when the inventory was frozen, is architecturally distinct from the candidates, and is 13.7 times slower than LIQE under the measured setup; it is therefore reserved as a reference model rather than screened as an on-line candidate.

DOVER is used only as an output-sensitivity reference and does not participate in screening [38]. The official `DOVER.pth` checkpoint and `dover.yml val-l1080p` views are used (Table S13). Its raw technical branch is primary; the uncalibrated technical–aesthetic sum is only a sensitivity analysis, and neither the official standardized fusion nor DOVER++ is computed. The same technical output is tested on the 800 constructed pairs, using a common 5-fps MP4 wrapper and paired fragment seed 42. Because DOVER was developed for user-generated video, its output is not treated as broadcast-domain human ground truth.

Score and target directions are declared before analysis. Direction-aligned correlation is $\rho^* = s_m s_y \rho$, where $s_m = +1$ for a higher-is-better method and -1 for a lower-is-better method, while $s_y = +1$ when a higher target means better quality and -1 when it means worse quality. All opinion targets used here—including KADID's released `dmos` field—are higher-is-better; degradation level is higher-is-worse. A negative ρ^* therefore denotes a genuine directional inversion rather than a scale convention.

Table 1 Scorers needed to interpret the main comparison. Sup.: K = released KonIQ-10k opinion-trained checkpoint; L = LIVE opinion training; OU = opinion- unaware; M = multi-database opinion mixture; – = no opinion data. Polarity is declared before evaluation. D = deployed, C = candidate, R = reference, X = diagnostic or ablation. Partial-coverage methods remain outside screening.

Method	Family	Measured construct	Sup.	Pol.	GPU	Role
SIFT+BoVW	handcrafted retrieval	TF-IDF-weighted visual-word count	–	↑	no	D
Keypoint density	handcrafted	detectable keypoints per frame	–	↑	no	X
DINOv2	self-supervised features	mean norm of patch tokens	–	↑	yes	X
Spatial Information	ITU-T complexity	Sobel energy	–	↑	no	X
BRISQUE	natural-scene statistics	deviation from learned natural statistics	L	↓	no	C
NIQE	natural-scene statistics	distance from a pristine model	OU	↓	no	C
DBCNN	CNN regression	bilinear distortion/content features	K	↑	yes	C
HyperIQA	CNN regression	content-adaptive quality regression	K	↑	yes	C
MUSIQ	transformer	multi-scale image quality	K	↑	yes	C
TOPIQ	transformer	top-down quality attention	K	↑	yes	C
CLIP-IQA	vision–language	antonym-prompt quality	OU	↑	yes	C
CLIP-IQA+	vision–language	learned-prompt quality	K	↑	yes	C
QualiCLIP	vision–language	degradation-ranked quality	OU	↑	yes	C
LIQE	vision–language	quality, distortion and scene	K	↑	yes	C
Q-Align	multimodal language model	discrete textual quality levels	M	↑	yes	R
CNN+BoVW (19 extractors)	descriptor ablation	TF-IDF-weighted visual-word count	–	↑	yes	X
ARNIQA; TReS	partial coverage	distortion manifold; ranking consistency	K	↑	yes	C

The 19-extractor ablation comprises ResNet-18/50/101, VGG-16/19, EfficientNet-B0/B4, MobileNet-V3-Large/V2, ConvNeXt-Tiny/Small, DenseNet-121/201, ResNeXt-50/101, Wide-ResNet-50/101, Inception-V3 and SqueezeNet-1.1. Released weights, endpoints and preprocessing are in Table S13.

3.3 Controlled degradation and constructed truth

KADID sensitivity is computed without treating derivatives of the same source as independent. For each distortion type, Spearman correlation between score and level is computed within each of the 81 references and averaged across references. Confidence intervals use 10^4 block-bootstrap resamples of the reference images with seed 42.

The broadcast experiment supplies a second controlled test. Eight degradations are applied at five levels: Gaussian blur, white noise, darkening, JPEG compression, low-bitrate H.264 transcoding, a 720p → 480p → 720p round trip, horizontal

motion blur and progressive desaturation. The last uses $S'_\ell = \mathrm{clip}(\alpha_\ell S, 0, 255)$ with $\alpha_\ell \in \{0.75, 0.55, 0.35, 0.18, 0.05\}$. H.264 is applied to the frame sequence; other transformations are frame-based.

Twenty of the 83 broadcast segments are sampled without replacement by a seeded simple random draw (seed 42); they represent 13 of the 19 operational topics (Table S2). Pairing each unmodified segment with every degradation and level gives $20 \times 8 \times 5 = 800$ binary decisions. The unmodified member is the construction-defined higher-fidelity signal, not a human perceptual judgment. Accuracy and paired differences are block-bootstrapped by segment (10^4 resamples, seed 42). A second design correlates score with degradation level within each segment–type pair. Overlay strata, the complete audit of predictions recorded in a commit whose metadata predates the experiment, and parameters are reported in Tables S2 and S12 and the reproducibility archive. Ties count as neither correct nor incorrect and remain in the denominator. A sensitivity analysis excludes darkening and desaturation, whose transformations define signal change more directly than universal perceptual worsening. These pairs test fidelity response, not preference between distinct broadcasts.

3.4 Screening and application inference

The method-screening rule is operational and was not preregistered. Sensitivity passes when the within-reference mean is at least .50 and at least 15 of 25 distortion types exceed .50. Generalization passes when the direction-aligned Spearman Rank Correlation Coefficient (SRCC) is at least .65 on *every* database absent from the released checkpoint's opinion training. The in-domain floor is confirmed when normalized agreement with Q-Align is above chance and the lower bound of its 95% topic-block bootstrap interval exceeds zero on the common 42 clusters. Cost is reported but not thresholded. Table S10 varies the numerical thresholds and the quantifier over unseen databases.

For cluster c with k_c candidates, independent uniform choices agree with probability $1/k_c$. The common-set floor is therefore $\bar{p}_e = |C|^{-1} \sum_{c \in C} 1/k_c = .470$. Normalized agreement is $\kappa_{\text{uniform}} = (p_o - \bar{p}_e)/(1 - \bar{p}_e)$. This is a custom uniform-choice correction, not Cohen's kappa. A positive point estimate is reported separately from a confirmed pass so that a wide interval crossing chance cannot be read as affirmative evidence.

External association uses Spearman correlation against MOS or DMOS. Confidence intervals (CIs) for KonIQ and CLIVE use the Fisher transform; KADID intervals and paired differences use reference-block bootstrap. Dependent correlations on the same images use Steiger's statistic [32]. The operational equivalence margin is $|\Delta\rho^*| < .02$, chosen as the largest association difference treated as practically negligible for this diagnostic rather than as a literature-wide standard; the paired difference and interval are also reported so stricter margins can be applied. Within each database, 630 pairwise Steiger contrasts cover 36 scores (17 methods and 19 ablations); Bonferroni correction gives $\alpha = .05/630$ and critical $|z| = 3.95$. Outside that family, the sole canonical-score–keypoint equivalence diagnostic uses two one-sided tests at $\alpha = .05$ against $\pm.02$; KADID also reports its paired block interval. The full accuracy, consistency and linearity triad after the monotonic mapping required by ITU-T P.1401 is reported in Table S4 [16].

For each operation, the application selects one representative per cluster. If y_c, l_c and h_c are the selected, lowest and highest reference scores in cluster c, define $\bar{y} = |C|^{-1} \sum_{c \in C} y_c$ and analogously $\bar{l}$ and $\bar{h}$. The reported global-achievable-range estimand is

$$100 \, \frac{\bar{y} - \bar{l}}{\bar{h} - \bar{l}}. \tag{3}$$

It is a ratio of means, not an average of independently rescaled clusters. Raw branch-score differences, mean per-cluster position, selected-maximum rate, range distributions and exclusions of the smallest 10% and 25% of ranges are reported in Table S11. Zero-range clusters are undefined for per-cluster position and are counted explicitly. The primary interval resamples the 19 topics and retains their clusters. A hierarchical sensitivity analysis additionally resamples clusters within each sampled topic. Both use 10^4 paired resamples and seed 42. Human compositional preference was not collected; the estimand is explicitly a model-reference output.

Experiments ran on one NVIDIA RTX 4060 GPU (8 GB). NR-IQA implementations use the PyIQA toolbox [7]; SIFT uses OpenCV and the vocabulary uses mini-batch K-means with $K = 300$ and `random_state=42`. Q-Align is quantized to 4-bit NormalFloat4 with double quantization and FP16 compute to fit the GPU [11]; the official PyIQA architecture supports fp16, 4-bit and 8-bit inference, with NF4 used in 4-bit mode. No numerical equivalence to fp16 is assumed; the reported eligibility set is therefore conditional on this documented NF4 reference configuration. Frames are decoded once and cached, and latency is re-measured in one software environment after ten warm-up and sixty timed iterations. The archive includes the scoring scripts, derived results, cluster definitions, annotations, seeds and environment information; redistribution of source news video remains subject to the broadcasters' terms. The final method inventory and analysis archive were frozen on 23 August 2026. The historical inference environment was not preserved as a fully pinned lockfile; the archive therefore reconstructs the reported statistics from hashed score caches and exposes the inference settings, but does not claim bit-identical regeneration of every neural score.

4 Results

4.1 Mechanistic diagnosis

The deployed inverse document frequency uses the vocabulary size, $\mathrm{idf}_{\mathrm{dep}}(v) = \log_{10}(K/\mathrm{df}(v))$, whereas canonical TF-IDF uses the number of documents, $\mathrm{idf}_{\mathrm{can}}(v) = \log_{10}(N/\mathrm{df}(v))$. Their relation is exact:

$$S_{\mathrm{dep}}(s) = S_{\mathrm{can}}(s) + D_s \log_{10}(K/N), \tag{4}$$

where D_s is the number of descriptors. The formulations therefore differ by an additive count term, not by rescaling. The identity holds within 3×10^{-11} on all 93 segment slots. At cluster scale $N \in \{2, 3, 4\}$, so the additional term rewards count and contributes a median 98.3% of the deployed score. On a database of approximately 10^4 documents, the term is negative and reverses the count ordering.

The more general factorization separates the two available channels:

$$\text{score}_i = \sum_v \text{tf}_{iv}\text{idf}_v = D_i \underbrace{\sum_v p_i(v)\text{idf}_v}_{\overline{\text{idf}}_i}. \tag{5}$$

Thus a score can vary through how many descriptors the extractor finds and through their mean IDF weight. The identity alone does not prove that either channel is uninformative; the relative contribution is measured empirically.

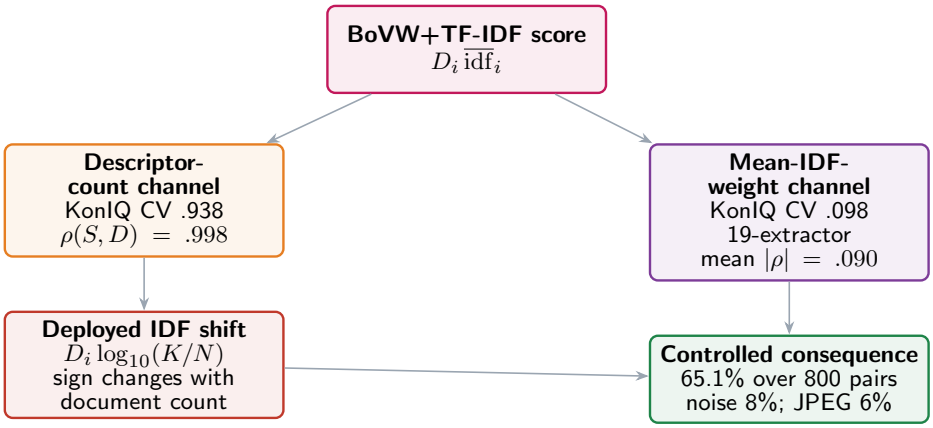


Fig. 2 Measured mechanism. The factorization identifies two channels; the data determine their relative variation and connect count dominance to the controlled-degradation failures.

On KonIQ-10k, the coefficient of variation is .938 for D_i and .098 for mean IDF weight. The canonical score consequently correlates .998 with descriptor count. Its association with human scores is statistically equivalent, within the operational $\pm.02$ margin, to that of keypoint density: $\rho^* = .447$ for both, with score-to-keypoint correlation .998. Steiger's $Z = .199$ is below the Bonferroni-corrected critical value of 3.95, and both one-sided equivalence tests reject. KADID gives the same association-level result under reference-block resampling: .125 for canonical BoVW and .123 for keypoint density, paired difference $+.0024$, 95% CI $[-.0019, +.0084]$. This is equivalence of measured MOS/DMOS association, not identity of the scores or constructs. The deployed formulation is almost exactly the sign-reversed count on the large databases ($-.9999$ on KonIQ, $-.9998$ on KADID and $-.9947$ on CLIVE). In the deployment regime its own descriptor count remains dominant, $\rho(S_{\text{dep}}, D) = +.994$.

Replacing SIFT with 19 convolutional extractors does not raise the external ceiling. Their mean $|\rho|$ is .090 on KonIQ and .112 on CLIVE, with no value above .238 on either database; KADID ranges from .156 to .418. Fixed spatial feature maps remove per-image count variation on the image databases, leaving the rarity channel measured by the ablation. On broadcast segments, however, the upstream key-frame selector varies the number of retained frames, so it reintroduces count variation: across the

19 variants, $\rho(S, D) = +.981$ to $+.997$. These results explain why exchanging the extractor changes individual choices without making the ordering a better predictor of quality. The explanation is limited to the evaluated vocabulary, normalization and extraction regimes.

Simple normalization does not repair the mechanism. Dividing by descriptor count exposes the weak mean-IDF-weight channel already measured by the ablation. Dividing by frame count is also mismatched to the deployed operation because its internal key-frame selector makes descriptor count nearly independent of duration. On the 42 clusters, frame normalization changes the uncalibrated DOVER two-branch-sum output only from 47.4% to 50.6% and induces an inverse duration association ($\rho = -.52$). Full normalization and pooling results are in Tables S6 and S9.

4.2 External validation and screening

Within-reference KADID sensitivity separates the deployed operation from specialized quality methods. Its mean direction-aligned correlation is $-.077$ (95% CI $[-.124, -.030]$), with only 5 of 25 types above .50. LIQE, QualiCLIP and CLIP-IQA+ reach .807, .849 and .808, respectively, each with 22 types above .50. Most remaining NR-IQA candidates also respond broadly, with means from .607 to .733.

Table 2 applies the same complete-case rule to every method. Released-weight exposure is visible because an in-database correlation is not a generalization result. The generalization column reports the lowest signed ρ^* over every unseen database, rather than using an absolute value that could hide inversion.

Table 2 Complete-case method screening. Sensitivity reports the within-reference mean and number of KADID types above .50. Generalization is the lowest signed ρ^* over all unseen databases. Floor is normalized agreement with Q-Align on the common 42 clusters and its topic-block lower bound (L). Sup.: K = KonIQ-10k opinion training; L = LIVE; OU = opinion-unaware; – = no opinion training; M = multi-database mixture. Q-Align is a reference, not a candidate.

Method	Sup.	Pol.	Sensitivity	Lowest unseen ρ^*	Floor; L	Eligible
LIQE	K	↑	.807; 22	.723	.551; .284	**yes**
QualiCLIP	OU	↑	.849; 22	.665	.371; .143	**yes**
CLIP-IQA+	K	↑	.808; 22	.654	.236; .034	**yes**
TOPIQ	K	↑	.721; 20	.511	.506; .289	no
MUSIQ	K	↑	.733; 20	.556	.551; .297	no
DBCNN	K	↑	.668; 19	.484	.461; .274	no
HyperIQA	K	↑	.673; 19	.462	.371; .122	no
NIQE	OU	↓	.607; 19	.368	.281; .012	no
BRISQUE	L	↓	.657; 19	.236	.371; .136	no
CLIP-IQA	OU	↑	.645; 20	.501	.371; .045	no
SIFT+BoVW (deployed)	–	↑	$-.077$; 5	$-.445$	.056; $-.308$	no
Q-Align	M	↑	.880; 23	–	–	reference

Three methods are eligible: LIQE, QualiCLIP and CLIP-IQA+. This is an eligibility set, not a ranking. QualiCLIP has the largest sensitivity mean, whereas LIQE

has the largest generalization margin. The set is unchanged when the generalization threshold varies from .60 to .65 or the sensitivity coverage cut varies from 12 to 22 types (Table S10); at .55 MUSIQ is added, and at .70 only LIQE remains. ARNIQA and TReS are reported on completed axes only, so neither enters this complete-case claim.

The database-level values show why the universal unseen-database clause matters. LIQE obtains $\rho^* = .723$ on KADID and .865 on CLIVE; QualiCLIP obtains .665 and .757; CLIP-IQA+ obtains .654 and .805. TOPIQ and MUSIQ are strong on the unseen authentic database (.811 and .789) but fall to .511 and .556 on KADID. DBCNN and HyperIQA similarly fall from .756 and .750 on CLIVE to .484 and .462 on KADID. An existential rule would admit methods whose favourable authentic result masks failure under controlled synthetic distortion. Conversely, the three eligible methods pass the floor with overlapping topic-block intervals, so the small in-domain sample cannot order them.

The floor axis illustrates that limitation directly. LIQE agrees with Q-Align on 32 of 42 clusters ($\kappa_{\text{uniform}} = .551$, 95% CI $[.284, .786]$); QualiCLIP agrees on 28 (.371, $[.143, .597]$); and CLIP-IQA+ on 25 (.236, $[.034, .435]$). The deployed operation agrees on 21, only 1.3 decisions above the uniform expectation, and its normalized interval crosses chance (.056, $[-.308, .412]$). The candidates pass the floor, but the broadly overlapping intervals do not support pairwise ordering. Results on the 15 human-annotated redundancy clusters are retained in Tables S7–S8 rather than mixed with the operational 42-cluster set.

The full P.1401 triad and database-level correlations are in Tables S4 and S15. The latter reports training-database contrasts descriptively, because database, content, architecture and objective are not matched. Conditioning on detectable keypoint density strongly attenuates the BoVW score but is not interpreted as a causal partition of "content density" (Table S16). Cost is also secondary to eligibility: in the common environment the deployed operation takes 182 s per cluster and LIQE 1.9 s (Table S14).

4.3 Constructed broadcast degradation

Table 3 removes the reference model from the decision: each pair contains the same content before and after a known degradation. The deployed operation is correct on 65.1% of 800 pairs. LIQE reaches 99.5%; the paired difference is +34.4 percentage points, segment-block 95% CI $[+31.1, +37.7]$. Every evaluated NR-IQA candidate exceeds the deployed score.

No method ties on these pairs. Excluding darkening and desaturation leaves 600 comparisons and does not repair the deployed operation: its accuracy is 63.7%, whereas LIQE reaches 99.8% and Q-Align 100% (Table S12). The main diagnosis therefore does not depend on treating those two transformations as universally worse in human perception.

The average conceals the diagnostic result. SIFT+BoVW is at or above 88% on five types but selects the degraded copy in 92% of white-noise pairs and 94% of JPEG pairs. Within segment, its score increases with severity for noise ($\rho = +.720$, CI

Table 3 Strict accuracy (%) on 800 unmodified–degraded broadcast pairs: 20 segments, eight types and five levels. The unmodified member is the construction-defined higher-fidelity signal; values below the 50% chance floor are systematic inversions. Ties remain in the denominator.

Method	dark	motion	blur	H.264	rescale	satur.	noise	JPEG	mean
SIFT+BoVW (deployed)	100	100	90	90	88	**39**	**8**	**6**	**65.1**
LIQE	97	100	100	100	100	100	99	100	**99.5**
TOPIQ	93	100	100	100	100	93	100	100	98.2
MUSIQ	93	100	100	100	100	86	100	100	97.4
QualiCLIP	99	100	95	100	96	76	96	71	91.6
ARNIQA	91	100	99	98	100	61	82	98	91.1
DBCNN	84	97	92	97	95	92	85	81	90.4
NIQE	100	100	98	97	98	**0**	100	96	86.1
CLIP-IQA+	76	99	79	100	91	87	84	64	85.0
HyperIQA	47	100	100	100	100	**30**	100	100	84.6
CLIP-IQA	58	100	83	100	92	91	55	76	81.9
BRISQUE	100	100	100	99	100	**0**	67	80	80.8

$[+.455, +.925]$) and JPEG ($+.640$, CI $[+.395, +.850]$). The mechanism is directly visible in the saved descriptor counts. The degraded copy has more descriptors in 92% of noise pairs (mean change +469) and 95% of JPEG pairs (mean change +289), with mean noise count increasing at every severity level. These measurements are consistent with noise and blocking creating detectable extrema that increase the count channel in Eq. 5; structure-removing distortions instead happen to move count in the direction of fidelity. The score is therefore responsive, but not consistently to quality (Table S12).

Q-Align is correct on 98.2% of these pairs (95% CI $[97.2, 99.1]$), which supports its technical-degradation competence on this broadcast content. It does not validate compositional preference. The exact downstream DOVER-T output reaches 97.6% ($[96.1, 99.0]$), or 99.8% ($[99.5, 100]$) without darkening and desaturation; neither result is a human judgment. The 19 extractor substitutions average 44.9% (CI $[41.0, 48.9]$) and remain dominated by their descriptor count, corroborating the mechanism while refuting the recorded prediction that fixed feature maps would force chance performance. The full prediction and overlay audits are in Table S12.

4.4 Pipeline substitution

LIQE is used for the direct operation substitution because it has the largest generalization margin among the three eligible methods and near-ceiling accuracy on the constructed pairs. With the same ten uniformly spaced frames and mean pooling, SIFT+BoVW and LIQE still differ in 16 of 42 clusters. The LIQE–SIFT DOVER-T contrast is +27.1 range points (topic CI $[5.6, 47.8]$; hierarchical $[2.4, 52.5]$) and +.0074 raw, with both intervals positive (Table S11). Table 4 reports the complete-operation comparison and all eligible methods; screening does not rank them.

Replacing the deployed visual-quality operation with the LIQE-based operation changes 22 of 42 representatives. On the primary DOVER technical branch, the global-achievable-range point estimate rises from 50.5% to 76.0%, a paired increase of +25.5 points. Its topic-block interval is $[-12.1, +54.9]$ and its hierarchical interval

Table 4 Model-reference output over the 42 operational redundancy clusters. Changed counts representatives differing from the deployed operation. DOVER-T is the raw DOVER technical branch and is not used in screening; Q-Align defines the in-domain screening floor. Both outputs use the global-achievable-range estimand in Eq. 3; Max is the percentage of clusters in which the selected representative has the highest DOVER-T score.

Quality operation	Changed	DOVER-T	Max	Q-Align
SIFT+BoVW	—	50.5%	54.8%	48.6%
LIQE	**22/42**	**76.0%**	59.5%	90.2%
QualiCLIP	19/42	71.9%	59.5%	80.6%
CLIP-IQA+	22/42	70.7%	50.0%	74.3%

is $[-17.2, +59.6]$; neither confirms a population-level effect. The raw branch-score difference is $+.0069$ (topic-block $[-.0028, +.0178]$; hierarchical $[-.0041, +.0197]$), while mean per-cluster position and selected-maximum rate increase by 7.2 and 4.8 points, respectively.

No cluster has zero DOVER-T range. Excluding the smallest 10% or 25% of ranges gives global-range differences of $+25.9$ and $+28.1$ points, so the positive point estimate is not generated solely by near-zero ranges. As a construct sensitivity analysis, the uncalibrated sum of DOVER's technical and aesthetic branches reproduces the earlier $+30.3$-point contrast, but that sum is neither the official standardized fused score nor a purely technical measure (Table S11).

The two model references have positive point estimates for the substitution but not on the ordering within the eligible set. That disagreement, together with overlapping intervals, is why the study reports eligibility rather than a winner. More importantly, neither column measures human preference for the composed summary. The supported end-to-end conclusion is that the operation substitution materially changes the selected evidence; the model-reference output effects are positive but imprecise, and improved viewer experience remains untested.

5 Discussion and limitations

The evidence converges on a localized diagnosis. The canonical score and keypoint density have statistically equivalent MOS/DMOS associations within the operational $\pm.02$ margin on KADID and KonIQ; the implementation adds a count-proportional term whose sign changes with collection size; and all 19 extractor substitutions remain bounded while tracking their descriptor counts. The constructed pairs reveal the consequence: noise and JPEG artefacts add detectable structure and reverse the ordering. A proxy can therefore appear competent on selected distortions while responding to the wrong construct.

This is not an impossibility theorem for all BoVW uses. Equation 5 identifies the two channels in this aggregation but does not prove that count or mean IDF weight can never correlate with quality under another detector, vocabulary, normalization

or distribution. The empirical result is that count dominates the evaluated configurations and that neither the measured mean-IDF-weight channel nor the extractor substitutions recover a reliable quality ordering.

The practical implication is to replace the operation rather than normalize it by segment length or insert a newer retrieval descriptor. LIQE, QualiCLIP and CLIP-IQA+ meet the common rule and run substantially faster than the deployed path, but the rule does not rank them. The LIQE substitution demonstrates propagation, not universal superiority. The complete operation changes 22 decisions; the matched control changes 16 and retains positive DOVER-T intervals. Thus scorer identity matters, but internal operation choices also contribute. DOVER does not isolate temporal modelling because architecture and training also differ.

The summarization claim is conditional. Cluster membership means that the pipeline treats segments as redundant alternatives; it does not establish identical completeness, narrative utility, audio quality or semantic relevance. A technically cleaner representative can omit context or carry worse audio, so visual quality remains a tie-breaker rather than an objective for the summary as a whole. A future human study should separately ask which segment has better technical visual quality and which is preferred in the composed summary.

The references also have bounded roles. KADID and the broadcast construction test responses to degradation, not preference between distinct recordings. Q-Align's 98.2% constructed-pair accuracy supports technical-degradation screening; the exact DOVER technical output used downstream reaches 97.6% on the same pairs. This validates its response to the constructed technical changes, not its use as human ground truth or a compositional preference model. The official standardized fusion and DOVER++ remain outside the analysis.

The application remains imprecise: 42 clusters are nested in only 19 topics, and both DOVER-technical intervals cross zero. Positive raw, range and rank point estimates, including after removing small-range clusters, do not confirm a population-level output effect. Coverage is also incomplete: ARNIQA and TReS are partial, MANIQA is absent, and later models may change the eligible set.

Finally, the auxiliary analyses are descriptive. KonIQ–CLIVE–KADID contrasts change content, capture, score distributions and model exposure together, while conditioning on keypoint density may remove legitimate quality signal. Tables S15 and S16 retain those diagnostics; the main conclusion rests on the exact factorization, dependency-aware external measurements, constructed pairs, matched-input control and controlled operation substitution.

6 Conclusion

This study audited the operation implementing a human-grounded visual-quality criterion in one multi-video summarization pipeline. The deployed SIFT+BoVW+TF-IDF score factors into descriptor count and mean IDF weight. Under the evaluated vocabularies, normalizations and extractors, descriptor count dominates: the canonical score and keypoint density have statistically equivalent MOS/DMOS associations within

the operational ±.02 margin on KADID and KonIQ, whereas the deployed formulation produces a near-exact sign-reversed count ordering in the large-database regime. In the broadcast domain, the deployed score tracks its own descriptor count at +.994 and remains bounded when SIFT is replaced by 19 convolutional extractors.

The mechanism predicts a consequential failure. Across 800 unmodified–degraded broadcast pairs, the deployed operation selects the construction-defined higher-fidelity member in 65.1% of decisions and systematically prefers white-noise and JPEG-degraded copies. LIQE reaches 99.5%. A dependency-aware screening rule retains LIQE, QualiCLIP and CLIP-IQA+; substituting the complete visual-quality operation with a LIQE-based operation, while holding every other pipeline stage fixed, changes 22 of 42 selected segments. The DOVER technical-branch global-range point estimate increases by 25.5 percentage points, but both topic-block and hierarchical intervals cross zero, so no population-level output effect is confirmed.

The supported conclusion is therefore specific: the audited operation is not a reliable perceptual-quality proxy in the tested regimes, and eligible NR-IQA replacements materially alter its decisions. With frames and pooling matched, SIFT+BoVW and LIQE still disagree in 16 of 42 clusters; 13 of these disagreements overlap the 22 changes in the complete-operation comparison. The study does not establish that all BoVW formulations fail, that redundancy clusters contain semantically identical alternatives, or that viewers prefer the modified summaries. Human pairwise evaluation and broader video-native quality assessment remain the next tests needed to connect technical visual quality to compositional value.

Statements and Declarations

Competing interests

The authors declare that they have no competing financial or non-financial interests that could have appeared to influence the work reported in this article.

Data availability

KADID-10k, KonIQ-10k and LIVE In the Wild are public. Supplementary Tables S1–S16 and a frozen archive with cache hashes, environment information, commands and a table-to-script manifest accompany the submission; these materials will receive a versioned Zenodo DOI before final acceptance. The HSMVideoSumm videos retain their original redistribution terms. A Git bundle preserves the prediction-recording commit and metadata, but not a third-party timestamp; hence "recorded in a commit whose metadata predates the experiment."

Code availability

Scripts and machine-readable results accompany the submission in the frozen private repository https://github.com/Prism411/evaluality-mtap-submission and will become public with the versioned archive after review.

Author contributions

J. L. S. G. designed and implemented the evaluation protocol, ran the experiments and wrote the manuscript. R. G. supervised the research, defined the problem framing within the multi-video summarization pipeline and revised the manuscript. Both authors read and approved the final version.

Ethics approval

Not applicable. The study uses publicly broadcast news material and three publicly released image quality databases; no new human subject data were collected.

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